Algebraic Curves
Chapter 8.1. Divisors

Let $\mathbb{C}$ be an irreducible projective curve, $f\colon \mathbb{X} \to \mathbb{C}$ the birational morphism from its nonsingular model $\mathbb{X}$, $\mathbb{K} = \Bbbk(\mathbb{C}) = \Bbbk(\mathbb{X})$ the function field, $\mathbf{p} \in \mathbb{C}$ identified with the places of $\mathbb{K}$, and $\mathrm{ord}_{\mathbf{p}}$ the order function at $\mathbf{p}$ in $\mathbb{K}$.

A 'divisor' D on $\mathbb{X}$ is a formal sum,

$$D = \sum_{\mathbf{p} \in \mathbb{X}} n_{\mathbf{p}} \mathbf{p}$$

Where $n_{\mathbf{p}} \in \mathbb{Z}$ is zero for all but a finite number of $\mathbf{p}$. The set of divisors on $\mathbb{Z}$ for an abelian group. Namely the free abelian group on the set of coordinates of $\mathbb{X}$. The 'degree' of a divisor is defined to the sum of its coefficient,

$$\deg D := \sum_{\mathbf{p}} n_{\mathbf{p}}$$

It is additive, i.e. $\deg [D + D'] = \deg D + \deg D'$. A divisor is said to be 'effective' or 'positive' if $n_{\mathbf{p}} \geq 0$ for all $\mathbf{p}$, and we write $\sum_{\mathbf{p}} n_{\mathbf{p}} \mathbf{p} = D \geq D' = \sum_{\mathbf{p}} m_{\mathbf{p}} \mathbf{p}$ if $n_{\mathbf{p}} \geq m_{\mathbf{p}}$ for all $\mathbf{p}$.

Let $\mathbb{C}$ be a plane curve of degree n and $\mathbb{G}$ another plane curve that does not have $\mathbb{C}$ as a component. We define the 'divisor' of $\mathbb{G}$ as,

$$\mathrm{div}[\mathbb{G}] := \sum_{\mathbf{p} \in \mathbb{X}} \mathrm{ord}_{\mathbf{p}}[\mathbb{G}] \mathbf{p} = \sum_{\mathbf{q} \in \mathbb{C}} \langle \mathbb{C}, \mathbb{G} \rangle [\mathbf{q}]$$

By Bezout's Theorem $\mathrm{div}\, \mathbb{G}$ is a divisor of degree mn where $m = \deg \mathbb{G}$. For any nonzero $z \in \mathbb{K}$ define its divisor,

$$\mathrm{div}[z] = \sum_{\mathbf{p} \in \mathbb{X}} \mathrm{ord}_{\mathbf{p}}[z] \mathbf{p}$$

Given that z has only a finite number of poles and zeroes, this is well-defined. In particular we may separately define,

$$(z)_0 = \sum_{\mathrm{ord}_{\mathbf{p}}[z] > 0} \mathrm{ord}_{\mathbf{p}}[z] \mathbf{p}, \qquad (z)_\infty = -\sum_{\mathrm{ord}_{\mathbf{p}}[z] < 0} \mathrm{ord}_{\mathbf{p}}[z] \mathbf{p}$$

Where $(z)_0$ is the 'divisors of zeroes' and $(z)_\infty$ is the 'divisors of poles' of z. They obey $\operatorname{div}[z] = (z)_0 - (z)_\infty$. We further have $\operatorname{div}[zz'] = \operatorname{div}[z] + \operatorname{div}[z']$ and $\operatorname{div}[z^{-1}] = -\operatorname{div}[z]$.

Proposition.
For all nonzero $z \in \mathbb{K}$, $\operatorname{div}[z]$ is a divisor of degree zero.

Proof.

Let $\mathbb{C}$ be a plane curve of degree n. Write $z = g/h$, where $g, h \in \Bbbk_h[\mathbb{C}]$ are forms of the same degree, namely the residue of forms $G, H \in \Bbbk[X, Y, Z]$ of degree m. Then,

$$\operatorname{div}[z] = \operatorname{div}[G] - \operatorname{div}[H] = mn - mn = 0$$

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The previous proposition implies that nonzero rational functions have the same number of poles and zeros when counted properly. Moreover $\operatorname{div}[z] \geq 0 \Leftrightarrow z \in \Bbbk \Leftrightarrow \operatorname{div}[z] = 0$. For the first implication $\operatorname{div}[z] \geq 0$ implies $z \in \mathscr{O}_{\mathbf{p}}[\mathbb{X}]$ for all $\mathbf{p}$. Let $z[\mathbf{p}_0] = \lambda_0$ for some $\mathbf{p}_0$, so that $z - \lambda_0$ has a root at $\mathbf{p}_0$. Being an element of $\mathbb{K}$ implies $\deg \operatorname{div}[z - \lambda_0] = 0$, but $z - \lambda_0$ creates an extra zero with $\operatorname{ord}_{\mathbf{p}}[z - \lambda_0] \geq \min\{\operatorname{ord}_{\mathbf{p}}[z], \operatorname{ord}_{\mathbf{p}}[\lambda_0]\} \geq 0$ still so no new poles are created so that $\deg \operatorname{div}[z - \lambda_0] > 0$, a contradiction. Necessarily $z - \lambda_0 = 0$, at which the rest follows. Moreover we also see that if $z, z' \in \mathbb{K} \setminus \{0\}$, then $\operatorname{div}[z] = \operatorname{div}[z']$ *iff* $z' = \lambda z$ for some $\lambda \in \Bbbk$ by converting products to sums and definiteness of div. We say that two divisors D, D' are 'linearly equivalent' if $D' = D + \operatorname{div}[z]$ for some $z \in \mathbb{K}$.

Proposition.
Linear equivalence is an equivalence relation. That is,

1. $D \sim 0$ *iff* $D = \operatorname{div}[z]$ for some $z \in \mathbb{K}$
2. If $D \sim D'$ then $\deg D = \deg D'$
3. If $D \sim D'$ and $D'' \sim D'''$, then $D + D'' \sim D' + D'''$
4. Let $\mathbb{C}$ be a plane curve. Then $D \sim D'$ *iff* there exists two curves $\mathbb{G}, \mathbb{G}'$ of the same degree s.t. $D + \operatorname{div}[\mathbb{G}] = D' + \operatorname{div}[\mathbb{G}']$.

Proof.

This is an exercise. The fifth point is given by the construction of $z = G/G'$ since then $\operatorname{div}[z] = \operatorname{div}[G] - \operatorname{div}[G']$.

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Noether's conditions translate well to divisors. Let $\mathbb{C}$ be a plane curve with only ordinary singularities. For each $\mathbf{q} \in \mathbb{X}$ denote $r_{\mathbf{q}} = m_{f[\mathbf{q}]}[\mathbb{C}]$ and define the divisor $E = \sum_{\mathbf{q} \in \mathbb{X}} (r_{\mathbf{q}} - 1)\mathbf{q}$. E is effective and has degree $\deg E = \sum_{\mathbf{p}} m_{\mathbf{p}}[\mathbb{C}](m_{\mathbf{p}}[\mathbb{C}] - 1)$. Any plane curve $\mathbb{G}$ s.t. $\operatorname{div}[\mathbb{G}] \geq E$ is called an 'adjoint' of $\mathbb{C}$. Question 7.19 says that a curve $\mathbb{G}$ is adjoint to $\mathbb{C}$ *iff* $m_{\mathbf{p}}[\mathbb{G}] \geq m_{\mathbf{p}}[\mathbb{C}] - 1$ for all singularities $\mathbf{p} \in \mathbb{C}$. Namely if $\mathbb{C}$ is nonsingular then all curves are adjoint.

Proposition.
(Residue Theorem) Let $\mathbb{C}$ and E be as defined. Let D and D' be effective divisors on $\mathbb{X}$ and $D \sim D'$. Let $\mathbb{G}$ be an adjoint of degree m s.t. $\operatorname{div}[\mathbb{G}] = D + E + A$ for some effective divisor A. Then there exists an adjoint $\mathbb{G}'$ s.t. $\operatorname{div}[\mathbb{G}'] = D' + E + A$.

Proof.

Let $\mathbb{H}$ and $\mathbb{H}'$ be curves of the same degree s.t. $D + \operatorname{div}[\mathbb{H}] = D' + \operatorname{div}[\mathbb{H}']$. Then,

$$\operatorname{div}[\mathrm{G}\,\mathrm{H}] = \operatorname{div}[\mathbb{H}'] + D' + E + A \geq \operatorname{div}[\mathbb{H}'] + E$$

Using Chapter 7.5. Proposition 3 on $\mathbb{C}$, $\mathbb{H}'$, and $\mathrm{G}\,\mathrm{H}$ we see that Noether's conditions are satisfied at all $\mathbf{p} \in \mathbb{C}$. Therefore by Noether's Theorem $\mathrm{G}\,\mathrm{H} = \mathrm{F}'\,\mathrm{F} + \mathrm{G}'\,\mathrm{H}'$ for some F', G' where $\deg \mathrm{G}' = m$. Hence,

$$\operatorname{div}[\mathbb{G}'] = \operatorname{div}[\mathrm{G}\,\mathrm{H}] - \operatorname{div}[\mathbb{H}'] = D' + E + A$$

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Question 8.1

Part A

Where $\mathbb{X} = \mathbb{C} = \mathbb{P}^1$, $\Bbbk(\mathbb{X}) = \Bbbk(t)$ for $t = X_1/X_2$, we compute $\operatorname{div}[t]$. Now $\operatorname{div}[X_1/X_2] = \operatorname{div}[X_1] - \operatorname{div}[X_2]$, so it suffices to compute these parts. Namely X_1 vanishes only at $[0:1]$ and X_2 vanishes only at $[1:0]$. At $[0:1]$ the local ring has uniformizing parameter X_1 as X_2 is invertible, so $\operatorname{ord}_{[0:1]}[X_1] = 1$, and vice-versa for X_2 at $[1:0]$. Hence,

$$\operatorname{div}[t] = 1[0:1] - 1[1:0]$$

Part B

Let us now compute $\operatorname{div}[\mathrm{f}/\mathrm{g}]$ where f and g are relatively prime in $\Bbbk[t]$,

$$\operatorname{div}[\mathrm{f}/\mathrm{g}] = \operatorname{div}[\mathrm{f}] - \operatorname{div}[\mathrm{g}] = \sum_{\mathbf{p}\in\mathbb{P}^1} \operatorname{ord}_{\mathbf{p}}[\mathrm{f}]\mathbf{p} - \sum_{\mathbf{p}\in\mathbb{P}^1} \operatorname{ord}_{\mathbf{p}}[\mathrm{g}]\mathbf{p}$$

Part C

Proposition.
Where $\mathbb{X} = \mathbb{C} = \mathbb{P}^1$, the degree of divisors of nonzero rational functions is zero.

Proof.

We compute,

$$\begin{aligned}
\deg \operatorname{div}[\mathrm{f}/\mathrm{g}] &= \deg\left[\sum_{\mathbf{p}\in\mathbb{P}^1} \operatorname{ord}_{\mathbf{p}}[\mathrm{f}]\mathbf{p} - \sum_{\mathbf{p}\in\mathbb{P}^1} \operatorname{ord}_{\mathbf{p}}[\mathrm{g}]\mathbf{p}\right] \\
&= \sum_{\mathbf{p}\in\mathbb{P}^1} \operatorname{ord}_{\mathbf{p}}[\mathrm{f}] - \sum_{\mathbf{p}\in\mathbb{P}^1} \operatorname{ord}_{\mathbf{p}}[\mathrm{g}] \\
&= \sum_{\mathbf{p}\in\mathbb{P}^1} \langle \mathbb{P}^1, \mathrm{f}\rangle[\mathbf{p}] - \sum_{\mathbf{p}\in\mathbb{P}^1} \langle \mathbb{P}^1, \mathrm{g}\rangle[\mathbf{p}] \\
&= n - n \\
&= 0
\end{aligned}$$

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Question 8.2

Where $\mathbb{X} = \mathbb{C} = \{Y^2Z - X(X-Z)(X-\lambda Z) = 0\} \subseteq \mathbb{P}^2$, $\lambda \notin \{0,1\}$, we compute $\operatorname{div}[X/Z]$ and $\operatorname{div}[Y/Z]$. To do so we first compute $\operatorname{div}[X]$, $\operatorname{div}[Y]$, and $\operatorname{div}[Z]$ separately. Note that $\mathbb{X} = \mathbb{C}$ is smooth, and our pullback is the identity.

First observe that the vanishing of X on $\mathbb{C}$ implies either Y or Z is zero, so $\{X = 0\} = \{[0:1:0],[0:0:1]\}$. For the first point, homogenizing in Y yields,

$$0 = Z - X^3 + (\lambda+1)X^2Z - \lambda XZ^2, \qquad \nabla \mathrm{f}_*[\mathbf{0}] = \hat{\mathbf{z}}$$

And so X is a suitable uniformizing parameter for $\mathscr{O}_{[0:1:0]}[\mathbb{X}]$, at which we may conclude $\operatorname{ord}_{[0:1:0]}[X] = 1$. Similarly at the second point, homogenizing in Z yields,

$$0 = Y^2 - X^3 + (\lambda+1)X^2 - \lambda X, \qquad \nabla \mathrm{f}_*[\mathbf{0}] = \lambda\hat{\mathbf{x}}$$

At which Y is a suitable uniformizing parameter for $\mathscr{O}_{[0:0:1]}[\mathbb{X}]$. From the lowest order term in X in our polynomial we thus see $\operatorname{ord}_{[0:0:1]}[X] = 2$. Therein,

$$\operatorname{div}[X] = 1[0:1:0] + 2[0:0:1]$$

Second observe that this vanishing of Y on $\mathbb{C}$ implies either $X = 0$, $X = Z$, or $X = \lambda Z$. These correspond to $[0:0:1]$, $[1:0:1]$, and $[\lambda:0:1]$ respectively. For the first point, homogenizing in Z yields,

$$0 = Y^2 - X^3 + (\lambda+1)X^2 - \lambda X, \qquad \nabla \mathrm{f}_*[\mathbf{0}] = \lambda\hat{\mathbf{x}}$$

At which Y is a suitable uniformizing parameter for $\mathscr{O}_{[0:0:1]}[\mathbb{X}]$, at which we may conclude $\operatorname{ord}_{[0:0:1]}[Y] = 1$. For the second point, homogenizing again in Z yields the same as previous, but,

$$\nabla \mathrm{f}_*[1,0] = (1-\lambda)\hat{\mathbf{x}}$$

So that Y is still a suitable uniformizing parameter here, so $\operatorname{ord}_{[1:0:1]}[Y] = 1$. At the last point we similarly compute,

$$\nabla \mathrm{f}_*[\lambda,0] = \lambda(\lambda-1)\hat{\mathbf{x}}$$

So that Y is still a suitable uniformizing parameter here, so $\operatorname{ord}_{[\lambda:0:1]}[Y] = 1$. Therefore,

$$\operatorname{div}[Y] = 1[0:0:1] + 1[1:0:1] + 1[\lambda:0:1]$$

Lastly observe that the vanishing of Z implies $X = 0$, so the only point of consideration in this case is $[0:1:0]$. Recall that homogenizing in Y yielded,

$$0 = Z - X^3 + (\lambda+1)X^2Z - \lambda XZ^2, \qquad \nabla \mathrm{f}_*[\mathbf{0}] = \hat{\mathbf{z}}$$

Whereby X was our uniformizing parameter. Then from the smallest terms in Z in our polynomial we see that $\mathrm{ord}_{[0:1:0]}[Z] = 3$. Therefore,

$$\mathrm{div}[Z] = 3[0:1:0]$$

Thus,

$$\mathrm{div}[X/Z] = -2[0:1:0] + 2[0:0:1]$$

$$\mathrm{div}[Y/Z] = 1[0:0:1] + 1[1:0:1] + 1[\lambda:0:1] - 3[0:1:0]$$

Question 8.3

Part A

Proposition.
Let $\mathbb{C} = \mathbb{X}$ be a nonsingular cubic. Let $\mathbf{p}, \mathbf{q} \in \mathbb{C}$. Then $\mathbf{p} \sim \mathbf{q}$ *iff* $\mathbf{p} = \mathbf{q}$.

Proof.

Backwards implication is immediate. For forwards implication note that the divisor E in the case $\mathbb{C}$ is simple is merely $E = 0$. Let ℓ be the unique projective line between $\mathbf{p}$ and $\mathbf{q}$. By Bezout's Theorem $\operatorname{div}[\ell] = 1\mathbf{p} + 1\mathbf{q} + 1\mathbf{k}$ for some point $\mathbf{k}$. Moreover ℓ is of degree one and $1\mathbf{p} + 1\mathbf{q}$ and $1\mathbf{k}$ are effective. Therefore by the Residue Theorem there exists ℓ' of degree one s.t. $\operatorname{div}[\ell'] = 2\mathbf{p} + \mathbf{k}$. Being of degree one ℓ' must be a line, which are simple, so cannot have $\operatorname{ord}_{\mathbf{p}}[\ell'] = 2$, a contradiction. Therefore $\mathbf{p} = \mathbf{q}$.

■

Part B

Proposition.
Let $\mathbf{p}, \mathbf{q}, \mathbf{r}, \mathbf{s} \in \mathbb{C}$. Then $1\mathbf{p} + 1\mathbf{q} \sim 1\mathbf{r} + 1\mathbf{s}$ *iff* the line between $\mathbf{p}$ and $\mathbf{q}$ intersects the line through $\mathbf{r}$ and $\mathbf{s}$ on a point on $\mathbb{C}$.

Proof.

Backwards implication follows directly by uniqueness of lines connecting two points. For forwards implication let ℓ be the line through $\mathbf{p}$ and $\mathbf{q}$ and ℓ' the line through $\mathbf{r}$ and $\mathbf{s}$. By Bezout's Theorem,

$$\operatorname{div}[\ell] = 1\mathbf{p} + 1\mathbf{q} + 1\mathbf{k}, \qquad \operatorname{div}[\ell'] = 1\mathbf{r} + 1\mathbf{s} + 1\mathbf{k}'$$

Namely by the Residue Theorem and the hypothesis applied to ℓ there must another line ℓ'' through $\mathbf{r}$, $\mathbf{s}$, and $\mathbf{k}$. As lines between two points are unique it follows that $\ell' = \ell''$, so that $\mathbf{k} = \mathbf{k}'$ which are points on $\mathbb{C}$ as desired.

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Part C

Proposition.
Fix $\mathbf{p}_0 \in \mathbb{C}$. Then $\mathbf{p} \oplus \mathbf{q} = \mathbf{r}$ *iff* $1\mathbf{p} + 1\mathbf{q} \sim 1\mathbf{r} + 1\mathbf{p}_0$.

Proof.

$(\Rightarrow)$

Recall that the addition $\oplus$ is defined as $\mathbf{p} \oplus \mathbf{q} = \varphi[\mathbf{p}_0, \varphi[\mathbf{p}, \mathbf{q}]]$, where φ outputs the third point on the line connecting its two parameters. Then $\mathbf{p} \oplus \mathbf{q} = \mathbf{r}$ means $\mathbf{r}$ is the third point

on the line ℓ' between $\mathbf{p}_0$ and the third point $\mathbf{k}$ on the line ℓ between $\mathbf{p}$ and $\mathbf{q}$. Namely,

$$\operatorname{div}[\ell] = 1\mathbf{p} + 1\mathbf{q} + 1\mathbf{k}, \qquad \operatorname{div}[\ell'] = 1\mathbf{p}_0 + 1\mathbf{k} + 1\mathbf{r}$$

Whereby $\mathbf{k} \in \mathbb{C}$. By Part B this implies $1\mathbf{p} + 1\mathbf{q} \sim 1\mathbf{r} + 1\mathbf{p}_0$.

$$(\Leftarrow)$$

Let $1\mathbf{p} + 1\mathbf{q} \sim 1\mathbf{r} + 1\mathbf{p}_0$. By Part B therefore there is a unique point $\mathbf{k}$ on $\mathbb{C}$ where the lines connecting the respective sides of the equation intersect. Hence,

$$\mathbf{p} \oplus \mathbf{q} = \varphi[\mathbf{p}_0, \varphi[\mathbf{p}, \mathbf{q}]] = \varphi[\mathbf{p}_0, \mathbf{k}] = \mathbf{r}$$

As desired.

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Question 8.4

Proposition.
For a cubic with a node $\mathbb{C}$, any two simple points $\mathbf{p}, \mathbf{q} \in \mathbb{C}$ are linearly equivalent.

Proof.

It suffices to prove that there exists $z \in \mathbb{K}$ s.t. $\operatorname{div}[z] = 1\mathbf{p} - 1\mathbf{q}$. Observe that $\mathbb{C}$ is projectively equivalent to $Y^3 + X^3 - XYZ$ whereby its only singularity is the node at $[0:0:1]$. Consider the unique projective lines ℓ, ℓ' between $[0:0:1]$, and $\mathbf{p}$ and $\mathbf{q}$ respectively. By Bezout's Theorem the sum of intersection numbers between $\mathbb{C}$ and ℓ is three. Being simple at $\mathbf{p}, \mathbf{q}$ we know the intersection number here is one. The tangents of $\mathbb{C}$ at $[0:0:1]$ are X and Y. Either of their vanishing means the other must also vanish, meaning that the only point left on $\mathbb{C}$ is $[0:0:1]$ which is the singularity. Hence no ℓ may ever be tangent to $\mathbb{C}$ at $[0:0:1]$. Hence the intersection number there is two. Thus ℓ and ℓ' intersect with $\mathbb{C}$ at $[0:0:1]$ and then $\mathbf{p}$ and $\mathbf{q}$ respectively, with,

$$\operatorname{div}[\ell] = 1\mathbf{p} + 2[0:0:1], \qquad \operatorname{div}[\ell'] = 1\mathbf{q} + 2[0:0:1]$$

Consider $\ell/\ell' \in \mathbb{K}$. By our computations,

$$\operatorname{div}[\ell/\ell'] = \operatorname{div}[\ell] - \operatorname{div}[\ell'] = 1\mathbf{p} + 2[0:0:1] - 1\mathbf{q} + 2[0:0:1] = 1\mathbf{p} - 1\mathbf{q}$$

As desired.

■

Question 8.5

Proposition.
Let $\mathbb{C}$ be a nonsingular quartic and $\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_2 \in \mathbb{C}$. Define $D = 1\mathbf{p}_1 + 1\mathbf{p}_2 + 1\mathbf{p}_3$. Let ℓ and ℓ' be s.t.,

$$\ell \bullet \mathrm{C} = 1\mathbf{p}_1 + 1\mathbf{p}_2 + 1\mathbf{p}_4 + 1\mathbf{p}_5, \qquad \ell' \bullet \mathrm{C} = 1\mathbf{p}_1 + 1\mathbf{p}_3 + 1\mathbf{p}_6 + 1\mathbf{p}_7$$

Where these points are all distinct. Then D is not linearly equivalent to any other effective divisor.

Proof.

Assume for contradiction D were linearly equivalent to some D'. Observe that $\ell\ell'$ is a conic (i.e. degree two curve) with,

$$\mathrm{div}[\ell\ell'] = \mathrm{div}[\ell] + \mathrm{div}[\ell'] = 2\mathbf{p}_1 + 1\mathbf{p}_2 + 1\mathbf{p}_4 + 1\mathbf{p}_5 + 1\mathbf{p}_3 + 1\mathbf{p}_6 + 1\mathbf{p}_7$$

In particular it is adjoint, so by the Residue Theorem there exists another conic F s.t.,

$$\mathrm{div}[\mathrm{F}] = D' + 1\mathbf{p}_1 + 1\mathbf{p}_4 + 1\mathbf{p}_5 + 1\mathbf{p}_6 + 1\mathbf{p}_7$$

Namely as $\mathbb{C}$ is nonsingular, $\mathrm{div}[\mathrm{F}] = \mathrm{F} \bullet \mathrm{C}$. Now degree d curves in the projective plane are uniquely determined by ${}_{d+2}\mathrm{C}_2 - 1$ points. For conics this means five points. By the intersection we are guarenteed $\mathbf{p}_1, \mathbf{p}_4, \mathbf{p}_5, \mathbf{p}_6, \mathbf{p}_7$, so the other points must be $\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3$. Hence $D' = 1\mathbf{p}_1 + 1\mathbf{p}_2 + 1\mathbf{p}_3 = D$.

■

Question 8.6

Part A

Proposition.
Let $\mathrm{D}[\mathbb{X}]$ be the group of divisors on $\mathbb{X}$, $\mathrm{D}_0[\mathbb{X}]$ the subgroup of divisors of degree zero, and $\mathrm{P}[\mathbb{X}]$ the subgroup of divisors of rational functions. Define $\mathrm{C}_0[\mathbb{X}] = \mathrm{D}_0[\mathbb{X}]/\mathrm{P}[\mathbb{X}]$ the divisor class group on $\mathbb{X}$. If $\mathbb{X} = \mathbb{P}^1$, then $\mathrm{C}_0[\mathbb{X}] = 0$.

Proof.

It suffices to prove $\mathrm{D}_0[\mathbb{X}] = \mathrm{P}[\mathbb{X}]$. Let $D = \sum_{\mathbf{p}} n_{\mathbf{p}}\mathbf{p} - \sum_{\mathbf{q}} m_{\mathbf{q}}\mathbf{q}$ be an arbitrary divisor of $\mathrm{D}_0[\mathbb{X}]$. On $\mathbb{P}^1$ we have the explicit construction,

$$z = \prod_{\mathbf{p}}(Y - \alpha_{\mathbf{p}}X)^{n_{\mathbf{p}}} / \prod_{\mathbf{q}}(Y - \beta_{\mathbf{q}}X)^{m_{\mathbf{q}}}$$

And being on $\mathbb{P}^1$, each term vanishes only at their respectively $\mathbf{p}$ or $\mathbf{q}$ with their specified order. Hence,

$$\mathrm{div}[z] = \sum_{\mathbf{p}} \mathrm{ord}_{\mathbf{p}}[z] = \sum_{\mathbf{p}} n_{\mathbf{p}}\mathbf{p} - \sum_{\mathbf{q}} m_{\mathbf{q}}\mathbf{q} = D$$

So $D \in \mathrm{P}[\mathbb{X}]$ and so $\mathrm{D}_0[\mathbb{X}] = \mathrm{P}[\mathbb{X}]$.

■

Part B

Proposition.
Let $\mathbb{X} = \mathbb{C}$ be a nonsingular cubic. Fix $\mathbf{p}_0$, defining $\oplus$ on $\mathbb{C}$. Then the map $\mathbf{p} \mapsto 1\mathbf{p} - 1\mathbf{p}_0$ is an isomorphism $(\mathbb{C}, \oplus) \cong \mathrm{C}_0[\mathbb{X}]$.

Proof.

Call this map ω. Let us first prove homomorphism. Using Question 8.13,

$$\omega[\mathbf{p} \oplus \mathbf{q}] = 1\mathbf{r} - 1\mathbf{p}_0 \sim 1\mathbf{p} + 1\mathbf{q} - 2\mathbf{p}_0 = \omega[\mathbf{p}] + \omega[\mathbf{q}]$$

For forwards injection let $1\mathbf{p} - 1\mathbf{p}_0 \sim 1\mathbf{q} - 1\mathbf{p}_0$, so $1\mathbf{p} \sim 1\mathbf{q}$. By Question 8.3. therefore $\mathbf{p} = \mathbf{q}$. For surjection we first characterize $\mathrm{C}_0[\mathbb{X}]$, namely we prove $\mathrm{C}_0[\mathbb{X}] = \{1\mathbf{p} - 1\mathbf{p}_0 + \mathrm{P}[\mathbb{X}] \mid \mathbf{p} \in \mathbb{C}\}$. Consider a line ℓ between $\mathbf{x}$ and $\mathbf{y}$. As both curves are simple,

$$\mathrm{div}[\ell] = 1\mathbf{x} + 1\mathbf{y} + 1\varphi[\mathbf{x}, \mathbf{y}]$$

Thus by writing $\mathrm{div}[\ell]/\mathrm{div}[\ell_0]$, where ℓ_0 is always the reference line, we are always able turn at least $1\mathbf{x} + 1\mathbf{y}$ to $-2\mathbf{p}_0 - 1\mathbf{r}_0$ at the cost of adding an $+1\varphi[\mathbf{x}, \mathbf{y}]$, but net still reducing the divisor by one. By inductively doing so we get that the divisor is of the form $1\mathbf{p} - 1\mathbf{p}_0$ for some $\mathbf{p}$ as desired. The surjection is then immediate.

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